

Lab 3: Inheritance and Polymorphism

Dr. Paweł Głaba

Season 2026

Basic Problem

In this exercise, create a base class called `Vehicle`. The class should contain the following fields:

- `brand`,
- `model`,
- `maxSpeed`.

Then implement three derived classes called `Car`, `Bike`, and `Truck`.

- The `Car` class should have an additional field `numberOfDoors` and a method that prints information about city driving.
- The `Bike` class should have an additional field `hasBasket` and a method that prints information about eco-friendly transport.
- The `Truck` class should have an additional field `loadCapacity` and a method that prints information about cargo transport.

Implement a common method in the parent class, for example `showInfo()`, and override it in each derived class. Make sure that you use encapsulation properly and override methods from the parent class where necessary. Write a console application called `VehicleDemo` that creates an array of six vehicle objects (two objects of each derived class) and iterates through the array to demonstrate inheritance and polymorphism.

Space Station Management

To model crew and operational diversity on the station, we now need to incorporate the concepts of inheritance and polymorphism.

Crew Members

Create at least three subclasses for different types of crew members that extend the parent `CrewMember` class. For example, you may use:

- `Engineer`,
- `Scientist`,
- `Pilot`,
- `Medic`.

Each subclass should have at least one additional field and one method specific to its role.

Officers

Implement three subclasses for different types of officers. For example:

- `CommandOfficer`,
- `MedicalOfficer`,
- `OperationsOfficer`.

All of them should extend the parent `Officer` class and contain at least one unique field or behavior.

AccessCard and PremiumAccessCard

Create a class called `AccessCard` with the following properties:

- access level,
- crew member name,
- issue date (use `LocalDate`).

Then create a class called `PremiumAccessCard` that extends `AccessCard` and adds extra properties such as:

- expiration date,
- special privileges.

Override the `toString()` method in both classes to display the card information in a readable format.

Remember: use inheritance only when there is a real “is-a” relationship between classes.